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SEMICONDUCTOR DEVICE

Abstract

A semiconductor device **100** includes a semiconductor chip **110,** a plurality of leads **102** arranged apart from each other around the semiconductor chip **110** in a plan view and extending from the semiconductor chip **110** side, an encapsulating resin **140** forming an outer shape such that at least lower surfaces and end faces of the plurality of leads **102** are respectively exposed, and a metal film **150** formed on a lower surface of the lead **102.** A lower surface of the lead **102** is formed at a position above a lower surface of the encapsulating resin **140.**

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION
[0001] This application claims the priority benefits of Japanese application no. 2024-027715, filed on Feb. 27, 2024, and Japanese application no. 2024-160017, filed on Sep. 17, 2024. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.
BACKGROUND
Technical Field
[0002] The present invention relates to a semiconductor device.
Description of Related Art
[0003] As electronic machines such as mobile phones and mobile machines become more functional, there is an increasing demand for miniaturization and thinning of semiconductor devices used in such electronic machines.
[0004] A common package for a semiconductor device formed by molding epoxy resin includes a structure in which a semiconductor chip is mounted on a die pad, which is part of a lead frame, and covered with epoxy resin to form an outer shape. One form of such a small package structure is a non-lead type DFN (Dual Flat Nonleaded) package.
[0005] In general, a DFN package is resin-encapsulated such that the lower surface of the lead is almost flush with the lower surface of the encapsulating resin, and the lower surface of the lead is exposed. Subsequently, a plating layer is formed on the lower surface of the lead, and the lead frame is singulated by cutting with a dicing device. The DFN package manufactured in this way functions with the lower surface of the lead, on which the plating layer is formed, as an outer lead.
[0006] As described above, in the case of a DFN package, when only the lower surface of the lead with the plating layer is bonded to the mounting substrate with solder or the like, it is difficult to secure sufficient bonding strength between the lead and the mounting substrate.
[0007] To suppress the decrease in bonding strength between the lead and the mounting substrate, for example, the invention described in Patent Document 1 (Japanese Patent Application Laid-Open (JP-A) No. 2000-294719) describes a semiconductor device in which an elongated groove-like recess is formed on the lower surface of the lead.
[0008] One aspect of the present invention provides a semiconductor device including a lead that may ensure bonding reliability with a mounting substrate even in a non-lead type package.
SUMMARY
[0009] A semiconductor device according to one embodiment of the present invention includes:
[0010] a semiconductor chip; [0011] a plurality of leads arranged apart from each other around the semiconductor chip in a plan view and extending from the semiconductor chip side; [0012] an encapsulating resin forming an outer shape such that at least lower surfaces and end faces of the plurality of leads are respectively exposed; and [0013] a metal film formed on a lower surface of the lead.
[0014] A lower surface of the lead is formed at a position above a lower surface of the encapsulating resin.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0015] FIG. **1** is a schematic top view (perspective view) of the semiconductor device according to the first embodiment of the present invention.

[0016] FIG. **2**A is a schematic side view (perspective view) of the semiconductor device illustrated in FIG. **1**.

[0017] FIG. **2**B is an enlarged view of part A in FIG. **2**A.

[0018] FIG. **3** is a diagram illustrating the manufacturing process of the semiconductor device according to the embodiment of the present invention.

[0019] FIG. **4** is a diagram illustrating the manufacturing process of the semiconductor device according to the embodiment of the present invention, continuing from FIG. **3**.

[0020] FIG. **5** is a diagram illustrating the manufacturing process of the semiconductor device according to the embodiment of the present invention, continuing from FIG. **4**.

[0021] FIG. **6**A is a schematic side view (perspective view) illustrating the semiconductor device according to the second embodiment of the present invention.

[0022] FIG. **6**B is an enlarged view of part B in FIG. **6**A.

[0023] FIG. **7**A is a schematic side view (perspective view) of a conventional semiconductor device.

[0024] FIG. **7**B is an enlarged view (schematic diagram) of part C in FIG. **7**A in the case where a conventional semiconductor device is bonded to a mounting substrate.

DESCRIPTION OF THE EMBODIMENTS

[0025] The embodiments for implementing the present invention are described in detail below with reference to the drawings. In addition, in the drawings, the same component parts may be given the same signs, and redundant explanations may be omitted.

[0026] Further, the X-axis, Y-axis and Z-axis shown in the drawings are assumed to be orthogonal to each other. The Z-axis direction may be referred to as “height direction” or “thickness direction.” The +Z-axis direction may be referred to as “upward”, and the −Z-axis direction may be referred to as “downward”. The surface on the +Z direction side of each component may be referred to as the “front surface” or “upper surface”, and the surface on the −Z direction side may be referred to as the “back surface” or “lower surface”. “Plan view” refers to viewing each component from the +Z direction side toward the −Z direction side. “Side view” refers to viewing each component transparently from the +Y direction side toward the −Y direction side.

[0027] Further, the drawings are schematic, and the width, depth, thickness ratio, etc. are not as shown. The quantity, position, shape, structure, size, etc. of each component are not limited to the embodiments shown below, but may be any quantity, position, shape, structure, size, etc. that is desirable in implementing the present invention.

[0028] FIG. **7**A is a schematic side view (perspective view) of a conventional semiconductor device. As illustrated in FIG. **7**A, in the conventional semiconductor device **900** of a DFN package, electrodes on the surface of a semiconductor chip **910** fixed to a die pad **901** with a conductive adhesive **920** are electrically connected to a plurality of leads **902** respectively by conductive wires **930**. A step portion **902***a* is formed at the corner portion between the lower surface **902***b* of the lead **902** and the end face **902***s* of the lead. An encapsulating resin **940** forms the outer shape of the semiconductor device **900** such that at least parts of the plurality of leads **902** are exposed. The die pad **901** and the plurality of leads **902** are formed of a metal material such as a copper alloy.

[0029] FIG. **7**B is an enlarged view (schematic diagram) of part C in FIG. **7**A in the case where a conventional semiconductor device is bonded to a mounting substrate. The lower surface **902***b* of the lead **902** is formed on the same plane as the lower surface **940***b* of the encapsulating resin **940**. A metal film **950** is formed on the lower surface **902***b* of the lead **902**. This metal film **950** is formed of a material with good solder wettability and is bonded to a mounting substrate **980** by solder **970**.

[0030] In conventional DFN packages as shown in FIG. **7**A and FIG. **7**B, it is common for the lower surface **902***b* of the lead to be formed almost on the same plane as the lower surface **940***b* of the encapsulating resin, and the interface between the lead **902** and the metal film **950** has a structure that is flush with the lower surface **940***b* of the encapsulating resin. Further, solder **970** is formed at the end portion of the interface between the lead **902** and the metal film **950**.
[0031] In such a structure, in response to conducting a reliability test that repeats temperature changes between high and low temperatures, stress generated due to differences in thermal expansion coefficients among the lead **902**, encapsulating resin **940**, and mounting substrate **980** concentrates at the interface between the lead **902** and the metal film **950**. This stress may cause cracks to occur at the interface between the lead **902** and the metal film **950**. The same applies to other non-lead type packages in the case where the lower surface of the lead and the lower surface of the encapsulating resin are formed almost on the same plane.
[0032] Thus, the semiconductor device according to one embodiment of the present invention forms the lower surface of the lead at a position above the lower surface of the encapsulating resin. As a result, it is possible to suppress the occurrence of cracks at the interface between the lead and the metal film, and provide a semiconductor device including a lead that may ensure bonding reliability with the mounting substrate even in a non-lead type package.
First Embodiment
[0033] FIG. **1** is a schematic top view (perspective view) of the semiconductor device according to the first embodiment of the present invention. As illustrated in FIG. **1**, the semiconductor device **100** according to the first embodiment of the present invention includes a semiconductor chip **110**, a plurality of leads **102**, and an encapsulating resin **140**.
[0034] The semiconductor chip **110** is a semiconductor chip for enabling the function of the semiconductor device **100**, and is mounted on the upper surface of the die pad **101** by being fixed with a conductive adhesive or the like. The plurality of leads **102** are arranged at a constant distance apart from each other on two opposing sides in the +X direction and-X direction of the die pad **101** in plan view.
[0035] The plurality of electrode pads (not shown) formed on the upper surface of the semiconductor chip **110** are electrically connected to the upper surfaces of the plurality of leads **102** by conductive wires **130**. The upper surface of the lead **102** is formed as a smooth surface. Thus, a wide area may be secured on the upper surface of the lead **102** for performing a second bond of the conductive wire **130**. This allows for securing the bonding area for wire bonding and may suppress the decrease in connection strength. The conductive wire **130** is, for example, a gold wire or a copper wire.
[0036] The hanging portions **103** extend from two sides of the die pad **101** in the +Y direction and −Y direction. It is noted that in this embodiment, the die pad **101**, the plurality of leads **102**, and the hanging portions **103** are formed of the same copper alloy material and have an integrated lead frame structure. FIG. **1** illustrates one combination that forms one semiconductor device. It is ultimately divided into individual semiconductor devices.
[0037] FIG. **2**A is a schematic side view (perspective view) of the semiconductor device illustrated in FIG. **1**. As illustrated in FIG. **2**A, the semiconductor chip **110** is mounted on the upper surface of the die pad **101** by being fixed with a conductive adhesive **120**. A metal film **150** is formed on the lower surface of the die pad **101** and is exposed from the encapsulating resin **140** to improve the heat dissipation of the semiconductor chip **110**. The periphery of the lower surface of the die pad **101** is formed to be thin. This creates a structure where the encapsulating resin **140** enters the lower surface around the die pad **101**, preventing the die pad **101** from falling out of the encapsulating resin **140**.
[0038] The side of the lead **102** near the die pad **101** is formed to be thin on the lower surface. This creates a structure where the encapsulating resin **140** enters the lower surface of the lead **102** on the die pad **101** side, preventing the lead **102** from falling out of the encapsulating resin **140**. The end

face **102***s* in the extending direction of the lead **102** is exposed from the side surface of the encapsulating resin **140**. A step portion **102***a* is formed at the corner portion between the lower surface **102***b* of the lead and the end face **102***s* of the lead.

[0039] The metal film **150** is continuously formed on the lower surface **102***b* and the step portion **102***a* of the lead **102**. The metal film **150** is exposed from the encapsulating resin **140** and functions as an external terminal.

[0040] FIG. **2**B is an enlarged view of part A in FIG. **2**A. As illustrated in FIG. **2**B, the lower surface **102***b* of the lead **102** is formed at a position above with a height t**1** from the lower surface **140***b* of the encapsulating resin **140**. The height t**1** is not particularly limited, but for example, it is 1 μm or more and 10 μm or less. The thickness t**2** of the metal film **150** is 10 μm or more and 50 μm or less, and the lower surface of the metal film **150** protrudes slightly from the lower surface of the encapsulating resin **140**. The thickness t**2** of the metal film **150** may be changed as appropriate. The material of the metal film **150** may be any material with good solder wettability, for example, lead, bismuth, tin, copper, silver, palladium, gold, or alloys of these may be mentioned.

[0041] As a result, the lower surface **102***b* of the lead **102** is formed at a position above the lower surface **140***b* of the encapsulating resin **140**, creating a structure where the stress generated due to the difference in thermal expansion coefficients between the encapsulating resin **140** and the mounting substrate does not concentrate at the interface between the lower surface **102***b* of the lead **102** and the metal film **150**. Thus, this semiconductor device **100** may suppress the occurrence of cracks at the interface between the lead **102** and the metal film **150**, and may ensure bonding reliability with the mounting substrate.

[0042] The manufacturing process of the semiconductor device **100** is described with reference to FIG. **3** to FIG. **5**.

[0043] As illustrated in FIG. **3**, the semiconductor chip **110** is fixed to the upper surface of the die pad **101** with conductive adhesive **120**. Subsequently, a plurality of electrode pads (not shown) formed on the upper surface of the semiconductor chip **110** are electrically connected to the upper surfaces of a plurality of leads **102** with conductive wires **130**. In this embodiment, the die pad **101** and the plurality of leads **102** are formed of the same copper alloy material and possess an integrated lead frame structure.

[0044] Next, as illustrated in FIG. **4**, the lead frame is held between two molds (not shown), and encapsulating resin is injected into the interior of the molds and solidified to encapsulate. The lower surface of the die pad **101** and the lower surface **102***b* of the lead **102** are exposed from the encapsulating resin **140**.

[0045] Subsequently, using a dicing device, a first cutting process is performed to form the step portion **102***a* of the lead **102** to a predetermined depth. The first cutting process cuts the depth t**4** of the step portion **102***a* from the lower surface **102***b* of the lead **102** to a thickness of 5% to 80% of the thickness t**3** of the lead **102**.

[0046] Next, by etching using a chemical solution, the lower surface **102***b* of the lead **102**, the step portion **102***a*, and the lower surface of the die pad **101** are formed at a position above the lower surface of the encapsulating resin. In the first cutting process, burrs of copper alloy, which is the material of the lead **102**, may occur on the step portion **102***a* and lower surface **102***b* of the lead **102**, and these burrs may cause short circuits between adjacent leads. Thus, by etching using a chemical solution, it is possible to remove these burrs and suppress the occurrence of short circuits between leads. Examples of the chemical solution include sulfuric acid peroxide and hydrochloric acid peroxide.

[0047] Subsequently, a metal film **150** is formed on the lower surface **102***b* of the lead **102**, the step portion **102***a*, and the lower surface of the die pad **101** exposed from the encapsulating resin **140** by electrolytic plating method.

[0048] Next, as illustrated in FIG. **5**, the semiconductor device **100** is manufactured by cutting and singulating the lead **102** and the encapsulating resin **140** at predetermined positions by a second cutting process using a dicing device. In this case, the blade width of the dicing device used in the second cutting process is narrower than the blade width used in the first cutting process. As a result, the end face **102***s* of the lead **102** is formed on the side surface of the encapsulating resin **140**.

[0049] In this manner, in the semiconductor device **100** of the present embodiment, the lower surface of the lead **102** is formed at a position above the lower surface of the encapsulating resin, which prevents the stress generated due to the difference in thermal expansion coefficients from concentrating at the interface between the lead **102** and the metal film **150**. As a result, the semiconductor device **100** may suppress the occurrence of cracks at the interface between the lead **102** and the metal film **150**, and may ensure bonding reliability with the mounting substrate even in a non-lead type package.

Second Embodiment

[0050] FIG. **6**A is a schematic side view (perspective view) illustrating the semiconductor device according to the second embodiment of the present invention, and FIG. **6**B is an enlarged view of part B in FIG. **6**A.

[0051] As illustrated in FIG. **6**A and FIG. **6**B, the semiconductor device **200** according to the second embodiment of the present invention is similar to the first embodiment of the present invention, except that a chamfering portion **202***r* is formed at the corner portion between the lower surface **202***b* and the step portion **202***a* of the lead **202**. Chamfering means removing the corner portion, and in the second embodiment, it is done as rounding processing to add roundness to the corner portion, but it is also possible to provide one or more angled surfaces at the corner portion.

[0052] In a structure having a corner portion between the lower surface **202***b* and the step portion **202***a* of the lead **202**, stress generated due to differences in thermal expansion coefficients concentrates at the corner portion. In the configuration shown in FIG. **6**A and FIG. **6**B, in the case of connecting the lead **202** and the mounting substrate, it is possible to suppress the concentration of stress on the corner portion, and it is also possible to improve the climbing of solder from the lower surface **202***b* of the lead **202** to the step portion **202***a*.

[0053] For this reason, the semiconductor device **200** according to the second embodiment of the present invention may ensure bonding reliability with the mounting substrate even in the case of a non-lead type package.

[0054] The present invention has been described with respect to one embodiment, but the present invention is not limited to this embodiment and includes designs within the scope that do not depart from the gist of the present invention.

[0055] For example, in this embodiment, the package of the semiconductor device is described as a DFN package, but it is not limited to this and may be other packages. The semiconductor chip is mounted on the upper surface of the die pad by adhering it with a conductive adhesive or the like, but the present invention is not limited to this and the semiconductor chip may be mounted on a resin film. The die pad may not exist. Further, the embodiment uses conductive wires for electrical connection between the lead and the semiconductor chip, but it may also be in a flip chip configuration. In the case of this flip chip configuration, the die pad no longer exists, and bumps formed on the upper surface of the semiconductor chip may be used for electrical connection between the lead and the semiconductor chip.

[0056] Further, the material of the lead frame is described as copper alloy, but it is not limited to this and may be other materials. The method of forming the metal film is described as electrolytic plating method, but electroless plating method may also be used.

Claims

1. A semiconductor device, comprising: a semiconductor chip; a plurality of leads arranged apart from each other around the semiconductor chip in a plan view and extending from the semiconductor chip side; an encapsulating resin forming an outer shape such that at least lower surfaces and end faces of the plurality of leads are respectively exposed; and a metal film formed on a lower surface of the lead; wherein a lower surface of the lead is formed at a position above a lower surface of the encapsulating resin.

2. The semiconductor device according to claim 1, wherein the lead comprises a step portion formed at a corner portion between a lower surface and an end face of the lead.

3. The semiconductor device according to claim 2, wherein the lead comprises a chamfering portion formed at a corner portion between a lower surface of the lead and the step portion.

4. The semiconductor device according to claim 3, wherein the chamfering portion is subjected to rounding processing.

5. The semiconductor device according to claim 3, wherein the chamfering portion is composed of a surface having an inclination.

6. The semiconductor device according to claim 2, wherein a height of the step portion is ½ or more of a thickness of the lead.

7. The semiconductor device according to claim 3, wherein a height of the step portion is ½ or more of a thickness of the lead.

8. The semiconductor device according to claim 4, wherein a height of the step portion is ½ or more of a thickness of the lead.

9. The semiconductor device according to claim 5, wherein a height of the step portion is ½ or more of a thickness of the lead.
